



UNIVERSITY of the
WESTERN CAPE



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA



UNIVERSITY OF
MPUMALANGA



UNIVERSITY OF
HOHENHEIM

UKUDLA Data Science Certificate

Explanatory Analysis

Markus Mößler

University of Hohenheim



UKUDLA

African German Centre

for Sustainable and Resilient Food
Systems and Applied Agricultural
and Food Data Science



science, technology
& innovation

Department:
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA



National
Research
Foundation



Supported by:
Federal Foreign Office



With funding from the:

Federal Ministry
of Research, Technology
and Space



With support from

Federal Ministry
of Agriculture, Food
and Regional Identity



Deutscher Akademischer Austauschdienst
German Academic Exchange Service

by decision of the
German Bundestag

Table of contents

1. Motivation
2. Concepts
3. Application
4. Key Message

Association is not yet a causal explanation

Descriptive question

- Do yield and precipitation vary together?
- How does the association differ by country or period?
- Which observations support the pattern?

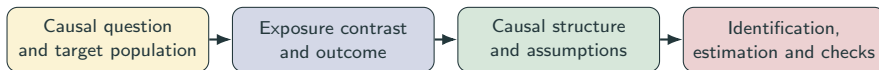
Causal question

- How would yield change under a different precipitation exposure?
- Compared with which alternative exposure?
- For which countries, seasons and years?

An observed correlation compares different observations. A causal effect compares potential outcomes for the same target units under alternative exposures.

A model can quantify an association; assumptions give it causal meaning.

Begin causal analysis before fitting a model



- Define what intervention or exposure difference the effect represents.
- State which variables occur before, during and after the exposure.
- Use domain knowledge to identify common causes and selection processes.
- Decide whether the available data can identify the intended effect.

Causal analysis is a research design supported by models—not a regression command.

Define the estimand before the estimator

Target quantity

- target population;
- exposure and contrast;
- outcome and time horizon; and
- aggregation level.

Potential outcomes

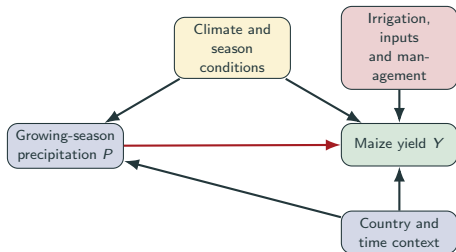
- $Y_i(p)$: yield under exposure p ;
- individual effect: $Y_i(p_1) - Y_i(p_0)$; and
- average effect over target units.

Fundamental problem

- one unit experiences one realized exposure;
- the alternative outcome is unobserved; and
- identification requires valid comparisons.

“An additional 100 mm” must specify timing, distribution within the season, affected area and the alternative condition. Otherwise the causal contrast is ambiguous.

A causal diagram makes assumptions inspectable



Use the diagram to ask

- Which variables cause both exposure and outcome?
- Which paths should adjustment block?
- Is a variable a mediator or collider?
- Which causes are measured adequately?

Arrows encode assumptions, not correlations discovered by software.

Regression estimates a conditional association

$$Y_{it} = \beta_0 + \beta_1 P_{it} + \alpha_i + f(t) + \varepsilon_{it}$$

Model components

- Y_{it} : an outcome for unit i in period t ;
- P_{it} : an exposure measured for that unit and period;
- α_i : country indicators;
- $f(t)$: chosen representation of time; and
- ε_{it} : unexplained model deviation.

Coefficient interpretation

- β_1 : modeled yield difference per precipitation unit;
- conditional on included country and time terms;
- linear and constant unless the model says otherwise;
- causal only when identification assumptions hold.

Adding controls changes the comparison; it does not automatically remove bias.

Identification requires assumptions beyond regression

Assumption	Meaning	Maize-project concern
Consistency	Observed outcome matches the specified exposure condition	Seasonal total hides timing and intensity
Exchangeability	No uncontrolled common causes of exposure and outcome	Temperature, irrigation, inputs and shocks
Positivity	Relevant exposure contrasts occur within adjustment groups	Limited overlap within countries and periods
No interference	One unit's exposure does not change another unit's outcome	Trade, shared water systems and regional shocks
Measurement	Exposure, outcome and confounders represent intended concepts	National yield and country-area precipitation

Identification connects the causal estimand to the observed-data distribution. If essential assumptions are implausible, a precise estimate can remain causally uninterpretable.

Diagnostics assess the model, not the causal design

Model checks can reveal

- nonlinear exposure-outcome patterns;
- changing residual variance;
- influential observations;
- temporal or grouped residual structure; and
- sensitivity to transformations and specification.

They cannot establish

- absence of unmeasured confounding;
- a well-defined intervention;
- correct causal direction;
- adequate measurement; or
- transferability beyond the target data.

Confidence intervals quantify sampling uncertainty under the model. They do not include unknown confounding, measurement error or misspecified causal structure.

Specify a causal analysis before estimation

Provisional target

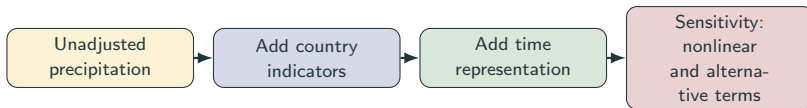
- units and target population;
- a well-defined exposure or intervention;
- outcome and timing;
- a meaningful exposure contrast; and
- target period and setting.

Required design work

- draw and justify a causal diagram;
- distinguish pre-exposure confounders from mediators;
- evaluate overlap within relevant groups and time;
- document unmeasured causes and exposure ambiguity; and
- decide whether the effect is identifiable.

Observed exposure variation is not automatically equivalent to a controlled intervention.

Compare estimates—then challenge their interpretation



- Report the target coefficient or contrast, uncertainty and effective n .
- Explain how the comparison changes across specifications.
- Inspect exposure overlap, residual plots, influence and temporal structure.
- Separate statistical precision from identification credibility.
- Label estimates as adjusted associations when causal assumptions remain unsupported.

Specification sensitivity is evidence to interpret—not a contest to find significance.

Key message: causal meaning comes from design

Reproducible artifacts

- causal question and estimand;
- causal diagram with justification;
- model specification table;
- coefficient and uncertainty table;
- diagnostics and sensitivity results; and
- code-linked explanatory report.

Conclusion must distinguish

- what the data describe;
- what the models estimate;
- which assumptions support identification;
- which assumptions remain doubtful; and
- which additional data or design are needed.

A credible causal analysis may conclude that the available data identify only adjusted associations. Making that boundary explicit is a scientific result, not a failure.

Estimate transparently—and earn the causal interpretation through design and assumptions.

Supported by

South African government and the National Research Foundation



science, technology
& innovation

Department:
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA



National
Research
Foundation

German government and the German Academic Exchange Service

Supported by:



Federal Foreign Office

With funding from the:



Federal Ministry
of Research, Technology
and Space

Supported by:



Federal Ministry
of Agriculture, Food
and Regional Identity

by decision of the
German Bundestag

Project Manager:



Federal Office
for Agriculture and Food



Deutscher Akademischer Austauschdienst
German Academic Exchange Service